OPIM639 · Innovations & Operations of Intelligent Online Marketplaces

Chagee: opening overseas stores right

Predict demand, provision the store, size the counter

Group 3

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Six chapters: one new store, four gates from forecast to review

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6	**Decide** Gate 4: review after opening, and what choosing well is worth	slides 8 and 9	1:50

About 7:00 on stage, then 3:00 for Q&A. Click a chapter to jump to it. Appendix slides are in the handout only.

One new store, no local sales history: four decisions before and after it opens

We are opening a store with no local sales history. What demand should we plan for, how much milk should we order, which counter should we install, and what should change after actual sales arrive?

Why it matters for Chagee

345 overseas stores in 7 markets and about 200 openings planned a year; each commits stock and a counter before local sales exist.

Proposed AI (not built here)

Estimate a new store's demand from comparable stores, cross-checked by pre-launch app sign-ups, and validate it against actual sales.

What this prototype calculates

Synthetic inputs and assumed relationships: a demand range, a milk-order calculation and a counter simulation. No trained model; membership data has not been shown to improve forecasts.

Who decides

People: teams choose the milk order and the counter from these outputs and local constraints.

INTERACTIVE EXPLAINER WHERE AI ENTERS A NEW STORE'S OPENING

Click a step. Today's practice is our reading of public sources and one industry conversation, not a Chagee statement.

1 Members → 2 Forecast → 3 Opening stock → 4 Counter → 5 After opening

STEP	TODAY (OUR READING)	WHAT AI ADDS	WHO DECIDES · WHERE
1 · Members	Members order through the app (our understanding).	**Read sign-ups and pre-orders as early demand evidence.**	Membership / CRM *slide 3*
2 · Forecast	Judgement-led by a local General Manager (our reading).	**A proposed demand range from member signals, with hype and habit separated by existing-market proxies.**	Market-entry leads and GMs *slide 3*
3 · Opening stock	Central, standardised supply of tea and milk.	**Size the first fresh-milk order to that range with newsvendor logic.**	Supply chain / planning *slide 4*
4 · Counter	Moving toward semi-automation: QR-scan dispensing, Chagee Express.	**Simulate the counter before the store exists, and flag which inputs would change the choice.**	Store format / equipment; franchisees *slides 5 to 7*
5 · After opening	Not covered by public sources.	**Compare actual demand with the range to narrow the next market's forecast.**	Market-entry leads *slide 8*

Printed with all five steps shown. Live: click a step to see one at a time.

FOUR GATES

Predict → Provision → Counter → Review.

Each gate states the decision, its evidence, a suggested action, the trade-off and what the model cannot establish. All figures are illustrative assumptions.

Context and problem

About 50 seconds. Start with one store: we are opening a store with no local sales history. What demand should we plan for, how much milk should we order, which counter should we install, and what should change once actual sales arrive? Chagee faces that about 200 times a year abroad. Be clear about three layers. The AI we propose would estimate a new store's demand from comparable stores, cross-checked by pre-launch app sign-ups, and be validated on actual sales; it is not built here. This prototype calculates from synthetic, assumed inputs: a demand range, a milk order and a counter simulation. People make the decisions. The deck walks the four gates: Predict, Provision, Counter, Review.

Sources: Scale facts and planned openings: L-jiemian. Direct sourcing, HQ distribution: L-prn (company release). Semi-automation: L-unsw, L-cyberrt. Kuala Lumpur opening day: L-gabgrowth (secondary). App ordering and GM-led forecasting: I-leeyen (INFORMAL - check with Lee Yen before presenting).

A closed app-to-store channel puts our problem in Topics 3 to 5, not the marketplace topics

INTERACTIVE EXPLAINER EIGHT COURSE TOPICS: CLICK ONE TO SEE WHY IT IS IN OR OUT

TOPIC 1
~~Introduction to Online Retailing~~
OUT
One closed channel, app to store: no online-versus-store trade-off to weigh (our reading).

TOPIC 2
~~Online Marketplaces~~
OUT
No third-party sellers, so no sell-versus-fulfilment-service choice (our reading).

TOPIC 3
Analytics and AI for Online Retailing
IN
→ Predict · slide 3
Predicting a new market's demand from app-member signals.

TOPIC 4
Supply Chain Management using AI and Robotics
IN
→ Provision · Perform
Supply-chain planning sizes the stock; equipment selection sizes the counter.

TOPIC 5
Order Fulfillment for Online Retailing
IN
→ Perform · Provision
Picking with humans and robots: the bucket brigade behind our counter model.

TOPIC 6
Last-mile Delivery for Online Retailing
NOT USED
Outside our question, which stops at the counter.

TOPIC 7
Service-oriented platforms
NOT USED
Outside our question.

TOPIC 8
~~Omni-channel Retailing~~
OUT
One channel, so no channel conflict; the franchise revenue share is out of scope.

INTERACTIVE EXPLAINER WHERE DOES UNCERTAINTY LAND?

Why this slice: with sourcing fixed, a demand miss lands in the store, on its opening stock or its counter speed (our reading, not a Chagee statement).

Rigid sourcing | Localised sourcing

Rigid sourcing Chagee today, our reading

Sourcing	○○○	Fixed: single origin
Distribution	○○○	Fixed: run from HQ
Opening stock	●●●	Absorbs the miss
Counter speed	●●●	Absorbs the miss

Localised sourcing For contrast only

Sourcing	●●○	Flexes with local suppliers
Distribution	●○○	Shorter, local
Opening stock	●○○	Less exposed
Counter speed	●●●	Still exposed

●○○ low ●●● high exposure to a demand miss

Printed: every tile shows its status and reason; both sourcing columns are shown.

OUR SCOPE

Three topics, three decisions.

Topic 3 feeds Predict; Topics 4 and 5 feed Provision and Perform. Sourcing is fixed by design, so we take it as given, and the franchise revenue share stays out of scope.

Scope: what we left out and why

About 40 seconds. The course has eight topics. Chagee runs one closed channel, from its app to its own stores, so the marketplace topics do not arise: there are no third-party sellers, no fulfilment-service choice and no channel conflict. Topics 6 and 7 are simply outside our question, which stops at the counter. Topics 3, 4 and 5 are where a new store's risk sits: predicting demand, planning the opening stock, and choosing people or robots at the counter. Use the toggle: because Chagee's sourcing is fixed, a demand miss can only be absorbed in the store. If asked why, open How the logic works: goods flow from suppliers through warehouses to stores, and a miss is absorbed wherever something can still change in time; with origin and routes fixed long before opening, only the store's opening stock and counter speed can react. That is why we chose this slice.

Sources: Topic names and content: OPIM639 course outline (AY2026-27), weekly lesson plans. Closed channel, no third-party sellers: brief.md section 3, our reading; app ordering rests on I-leeyen (INFORMAL). Direct sourcing and HQ distribution: L-prn. Franchise revenue share: out of scope, no figure claimed.

Predict: plan on a demand range, not one number, and treat the range as an assumption to test

GATE 1 · PREDICT

DECISION
What demand should opening-stock and counter planning assume?

SUGGESTED ACTION
Plan on about 3,889 cups a week, within an illustrative 80% range of 2,625 to 5,153; treat it as an assumption to test.

EVIDENCE
Base **3,889 cups/week** 80% range **2,625–5,153**
Half-width **±32.5%**

TRADE-OFF
Plan too low: shortages, queues and lost sales. Plan too high: wasted fresh milk and machines or staff that sit idle.

LIMITATION
The range is assumed, not calibrated, and demand can fall outside it. It cannot approve a site or show that a store is viable.

Demand prediction `COURSE CONCEPT · TOPIC 3` Estimate how much customers will buy, as a range rather than one number. *Informs: the demand assumptions for stock and counter planning*

Proposed AI: comparable stores first, sign-ups as a cross-check

A model trained on Chagee's existing stores would set the base; app sign-ups would cross-check it. Not built here: our base is assumed.

A narrower range is an assumption to test

The slider narrows it from ±50% to ±15% by assumption; no evidence yet that member data does this.

Hype versus habit needs a proxy

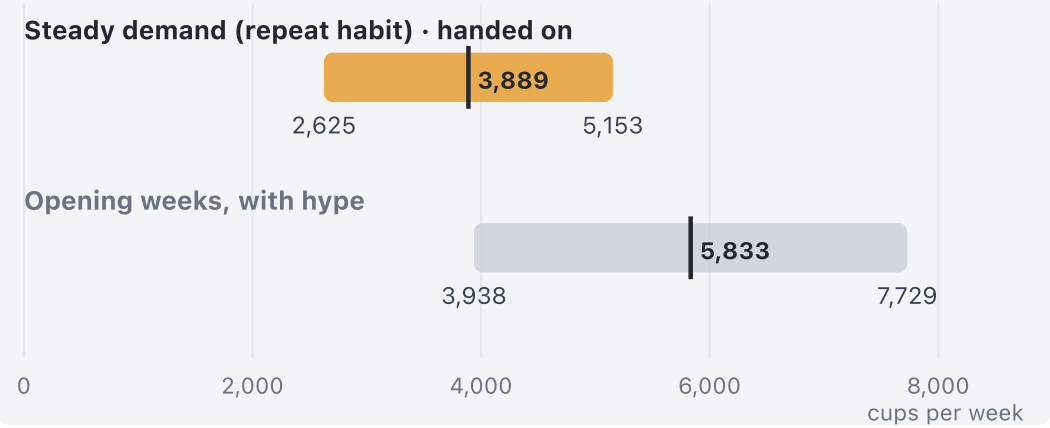
No repeat history yet: use repeat rates in existing markets or nearby stores. Kuala Lumpur reportedly sold 1,000+ cups on day one (secondary).

Blind spot: tourists and non-app orders

They sit outside the member signal, so the range runs low where tourists matter.

INTERACTIVE EXPLAINER DEMAND RANGE BEFORE OPENING: A SENSITIVITY

Weekly cups at one new store. The bar is an illustrative 80% demand range: we assume about that share of outcomes falls inside it.



Assumed signal strength — 50% → ±32.5%

Opening hype removed | Hype included

Plan on about 3,889 cups a week of steady demand (opening hype removed). Illustrative 80% demand range: 2,625 to 5,153. The range is assumed, not calibrated; demand can fall outside it.

Printed at the default assumed signal strength (50%) with opening hype removed.

WHAT THE RANGE IS FOR

Low end (2,625)
Risk of excess fresh stock and unused capacity.

High end (5,153)
Risk of shortages, queues and lost sales.

Provision uses the range with shortage and waste costs to choose the milk order.

Counter uses demand, uncertainty and local wages and machine costs together.

Not in the signal: tourists and non-app orders.

Predict

About 40 seconds; under time pressure, read only the gate card. Chagee orders go through the app (our understanding, to be confirmed), so a pre-launch waitlist could be read as a demand signal. Be clear what the slider is: we assumed that more signal narrows the band, along a straight line between two end points; we have not shown member data does that. Separating opening hype from repeat habit also needs a proxy, because a new store has no repeat history: repeat rates in Chagee's existing overseas markets, nearby stores, or waitlist members who already order elsewhere. So this is a method to back-test on past openings. Gate card: does the band change the counter? Barely: mixed overtakes all-human only beyond about ±57%. The band matters most for stock, the next slide.

Sources: App-only ordering with tourist exceptions: I-leeyen (INFORMAL - check before presenting). Kuala Lumpur opening day: L-gabgrowth (secondary). Existing overseas markets: L-jiemian (7 markets). Not novel - AI forecasting and siting: L-starbucks. Band narrowing with signal: assumption (predict.signal_strength_default, demand.uncertainty_no_signal / _strong_signal). Trigger: sim/ breakEven over the band half-width, default volume, moderate surge.

Provision: size the first milk delivery by weighing unused milk against running short

GATE 2 · PROVISION

DECISION

How much fresh milk should the first delivery bring?

SUGGESTED ACTION

Order about 788 L for the first 7 days, using steady demand (opening hype removed).

EVIDENCE

Short vs waste, per L **S$25.47 vs S$2.55**

Order covers **91% of outcomes** Expected waste **209 L**

Expected shortage **42 cups**

TRADE-OFF

Order less: more missed cups at S$3.82 each. Order more: more milk thrown away at S$2.55 a litre. A shortage costs more, so the order sits above base demand.

LIMITATION

Still runs short in about 9% of outcomes. It sizes the first delivery only; later deliveries should follow actual sales. Milk costs are generic.

Newsvendor logic Choose the order quantity by balancing the cost of unused stock against the cost of running short. *Informs: the size of the first milk delivery*

Preparing inventory for fulfilment COURSE CONCEPT · TOPIC 5 Hold the right stock before the orders arrive. *Informs: what the store holds before its first customers*

Two ways to be wrong

Too much fresh milk expires; too little turns customers away. Newsvendor logic: order until one more litre's expected waste cost equals the expected lost-sale cost it avoids.

Here a stockout costs more than waste

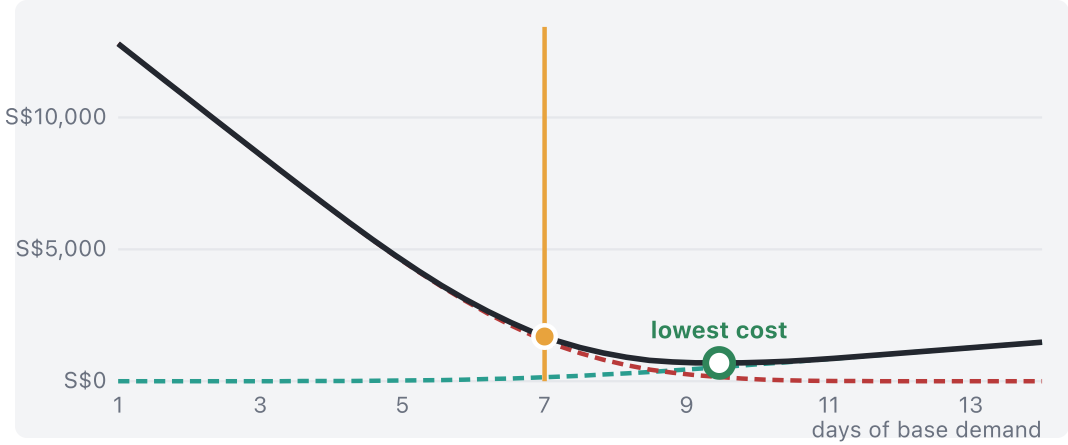
A missed cup loses S$3.82; a spare litre of milk wastes S$2.55. So the best order sits high in the band, at the 91% point of demand.

Shelf life caps the buffer

Fresh milk keeps about 7 days (illustrative), so extra stock cannot be carried forward. A narrower band from slide 3 is the cheaper fix.

INTERACTIVE EXPLAINER **OPENING STOCK CALCULATOR**

Expected cost of the first delivery at one store, using the range from slide 3. Milk costs are generic: overseas sourcing is not public.



Days of opening stock 7 days

Order about 788 L for the first 7 days, using steady demand (opening hype removed). That is 9.5 days of base demand, just above the top of the range (top of range 9.3 days).

The opening weeks are forecast higher (hype ×1.5); if that hype is real, this first delivery runs short: include it on slide 3 to size for it.

Printed at the band from slide 3 and the default of 7 days of stock.

AT 7 DAYS OF STOCK

S$1,695 expected

Milk wasted **S$149**

Cups missed **S$1,546**

BEST ORDER AT THIS BAND (±32.5%)

9.5 days **S$694**

Same order, strong signal **S$320**

Hand-off: open with 788 L of milk. The counter is decided by volume and input costs, not this band (slide 7).

Provision

About 30 seconds; under time pressure, read only the gate card. Fresh milk is a newsvendor problem: order until one more litre's expected waste cost equals the expected lost-sale cost it avoids. A lost cup costs far more than a wasted litre, so the best order sits above the base forecast, and that holds unless waste costs over S$25.47 a litre. A wider band raises the cost of even the best order: that is the pre-launch signal's value in stock terms. Caveat: milk costs are generic; Chagee's overseas sourcing is not public.

Sources: All figures: assumptions.json (illustrative). Milk trigger: sim/ milkTrigger, stockout cost per cup / milk per cup, against the approved milk waste range. Overseas milk sourcing: unknown, deliberately not claimed (brief.md section 9).

In a surge, people pass work along like a bucket brigade; machines cannot, so the slowest station sets the pace

Bucket brigade `COURSE CONCEPT · TOPIC 5` Workers pass each order along a line and rebalance themselves, slowest first. *Informs: why people absorb a surge*

Bottleneck `COURSE CONCEPT · TOPIC 5` The slowest step sets the pace of the whole line. *Informs: why fixed machines queue at the slowest step*

People pass the work along

Each worker carries a drink downstream and hands it over. In a surge the handover point moves, so the load evens out: the Week 5 bucket-brigade result.

Machines stay at their station

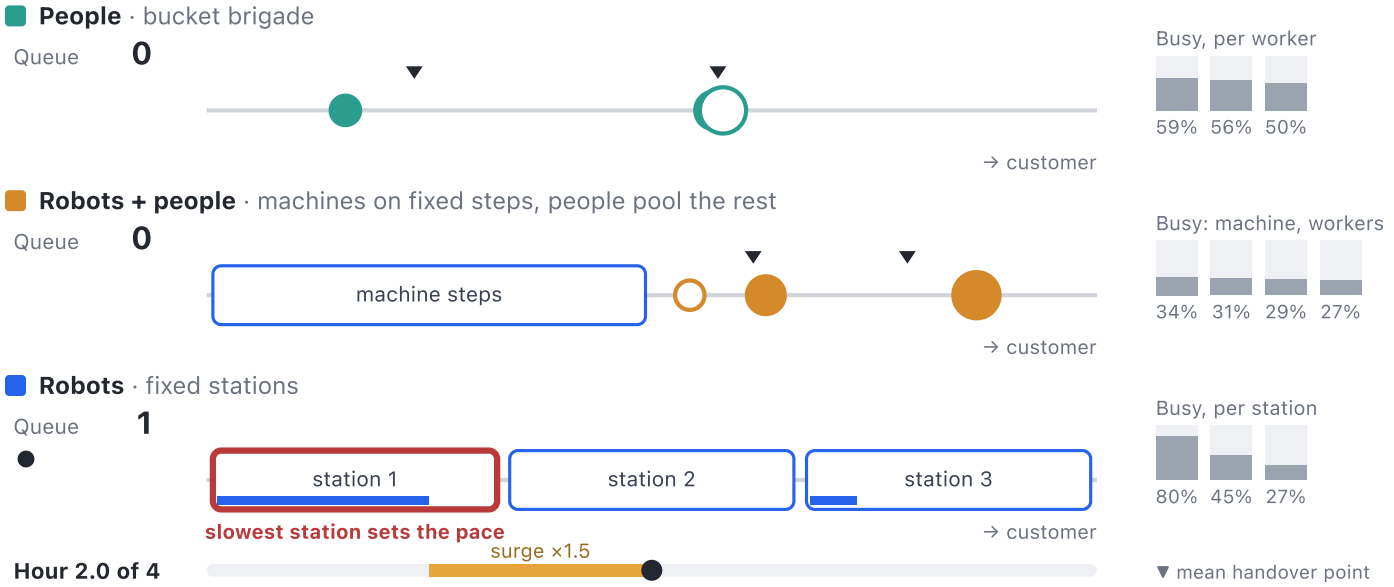
A fixed station cannot hand work back or take on more. The queue builds at the slowest one, wherever it sits on the line.

Sized for the peak, machines sit idle

Switch to as sized in our cost model: the robots stop queuing, but they are busy only part of the time. Idle capacity is what a store pays for.

INTERACTIVE EXPLAINER INSIDE THE STORE: A BUCKET BRIGADE AT THE COUNTER

A movement illustration in the Week 5 units, not a cost model and not measured Chagee service times. Times and costs come from the next two slides.



Moderate ×1.5 surge, 4-hour run	Machines matched to crew		As sized in cost model	
	Peak queue	Busy (range)	Peak queue	Busy (range)
People	0	57%–66%	0	57%–66%
Robots + people	0	28%–32%	0	10%–33%
Robots	2	38%–90%	1	19%–40%

Printed at the end of the surge window; the table compares both sizing settings over the full run.

OUR VIEW

People flex; machines come in whole units.

So the counter choice turns on volume. The next two slides measure it: time to serve, then cost.

SPEAKER NOTES · PAGE 7

Perform

About 35 seconds. Press play. Top lane: people work as a bucket brigade. When the surge hits, the faster worker takes over more of each drink, the handover point moves and the load evens out. Bottom lane: robots are fixed at their station, so the queue builds at the slowest one and nobody can take its work. Flip to as sized in our cost model: the robots stop queuing, but they sit idle much of the run, and idle machines are what you pay for. This is a mechanism, in the Week 5 units; the next slide puts it in a real store layout in seconds.

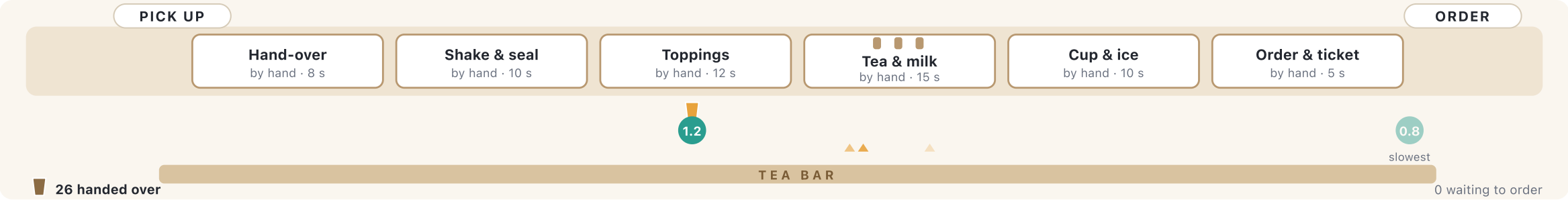
Sources: Bucket-brigade engine: course Week 5 explainer (reference, read-only), wrapped by sim/mechanism.js. As-sized ratios read from sim/ peakUtilisation and plan. Illustration settings: assumptions.json illustration (mixed_sizing needs approval).

In the peak hour, robots hand over a cup in about 49 seconds; a counter like today's takes about 2.6 minutes

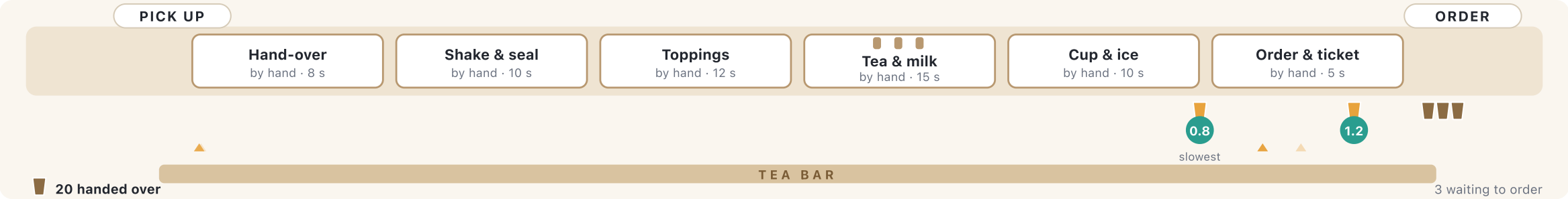
INTERACTIVE EXPLAINER STORE REPLICA: TWO COUNTERS IN THE PEAK HOUR, SIDE BY SIDE

Illustrative layout after a Chagee tea bar: order right, brewing behind the counter, pick-up left. Choose each counter; people move as the Week 5 bucket brigade; staff and arrivals from the cost model.

TOP **People, slowest first (Week 5)** — Minute 20 · Handovers 30 · 0.8× working 79% · held back 0% · 1.2× working 88% · held back 0%



BOTTOM **People, fastest first** — Minute 20 · Handovers 28 · 1.2× working 88% · held back 54% · 0.8× working 87% · held back 0%



Number = worker speed · cup above = drink in hand · faded = walking back · red ring = held back by the colleague ahead · ▲ = handover. Robots stay put; cups move. Both counters see the same customers.

Printed at minute 20 of the peak hour, moderate surge: people slowest first (top) against fastest first (bottom).

Time to serve, cost model (median · slowest tenth · lost):

Today's counter **2:34** · 5:03 · 2% lost — All-human **2:42** · 5:10 · 2% lost — All-robot **0:49** · 1:19 · 0% lost

Observed: not measured yet

OUR VIEW

Robots buy speed in the peak; whether it pays is the next slide.

Perform

About 45 seconds. Two counters laid out like a Chagee tea bar, seeing the same customers: order on the right, brewing behind the counter, pick-up on the left. It plays on its own. Top: people as the Week 5 bucket brigade, slowest first. Each worker carries a cup towards pick-up; the last hands it over, walks back and takes over from the colleague behind, so the line balances itself. Bottom: the same people, fastest first. The fast worker keeps catching the slow one and is held back, the red ring, and fewer cups get out. Switch the bottom to All-robot: nobody moves, the cups move from machine to machine. The times come from the cost model, in the line underneath: today's counter hands over a cup in about 2.6 minutes at the median, an all-robot counter in about 49 seconds; open it for the chart. None of these is measured. The next slide puts a price on each counter.

Sources: Semi-automation (machine dispenses the formulation): L-unsw; Chagee Express semi-automated machine: L-cyberrt. Layout: illustrative, after a Chagee tea-bar storefront (order right, pick-up left). Animation: sim/store.js (Week 5 brigade rules, illustration.walk_back_speed). Times and lost orders: sim.js serveTimes, a read-out of the cost model's peak hour. No measured serve time exists in our sources; the informal sub-one-minute target (I-leeyen) is not used.

Counter: at our inputs all-human and mixed are S$0.02 a cup apart, so choose with all inputs together

GATE 3 · COUNTER

DECISION

Which counter should this store install: all-human, mixed, or full automation?

SUGGESTED ACTION

At 556 cups a day all-human is cheaper by S$0.02 a cup: close, but above the S$0.01 tie line. Choose using all current inputs together.

EVIDENCE

Cost per cup **all-human S$0.65 · mixed S$0.66 · all-robot S$0.91**

All-robot machines busy **21% of open hours**

TRADE-OFF

All-human: cheapest here, but depends on trained staff. Machines: faster in the peak, but idle and costly when volume is low.

LIMITATION

Small gap on a simplified staffing model that misses its Week 5 check (4.7% and 6.8% against ±3%). No measured Chagee service times.

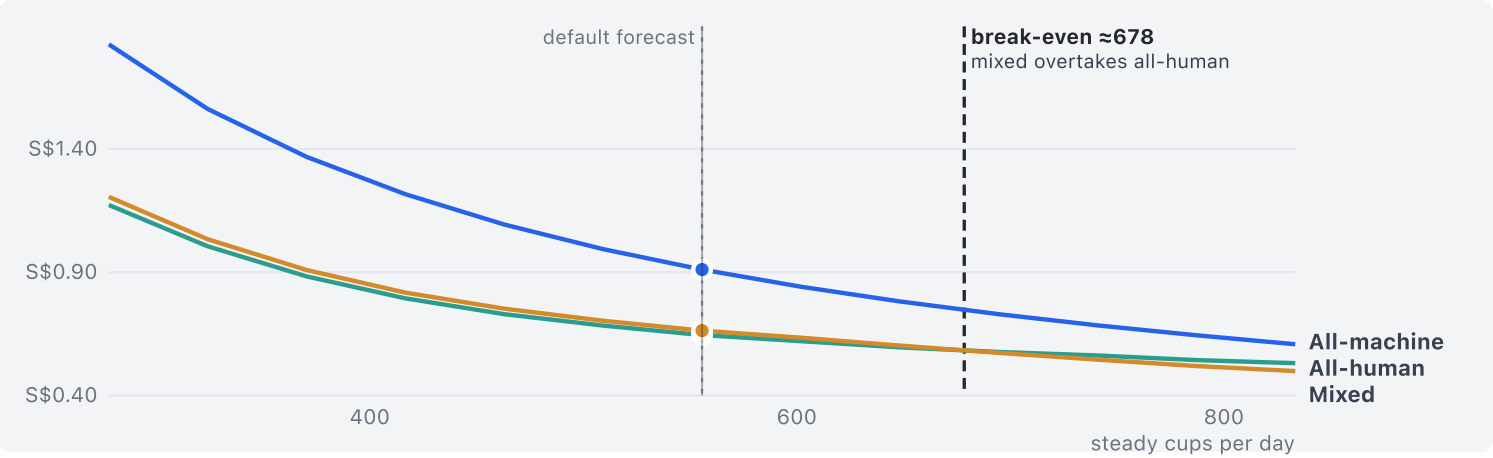
INTERACTIVE EXPLAINER COST PER CUP, BY COUNTER

Faster service does not mean a lower cost per cup: at low volume, machine capacity sits unused. Labour, machines and lost sales per cup served, by steady volume.

Capacity utilisation The share of available capacity that is actually used. *Informs: why idle machines raise the cost per cup*

Selecting robotics equipment COURSE CONCEPT · TOPIC 4 Choose automation that fits the volume and the work, not the most automation. *Informs: which counter to install*

All-human Mixed All-machine



All-human	Mixed	All-machine
S$0.65	**S$0.66**	**S$0.91**
cheapest	+S$0.02	+S$0.27

At 556 cups/day. All-robot machines are busy 21% of open hours.

Printed at the default forecast.

ONE-VARIABLE THRESHOLDS: WHICH ASSUMPTIONS DESERVE A CLOSER LOOK

MIXED WINS WHEN…	IS	OURS	GAP
Machine cost	below **S$1,800/month**	S$1,909/month	-6%
All-human coordination	below **0.85**	0.90	-6%
Wage per hour	above **S$15.97/h**	S$15.06/h	+6%
Mixed coordination	above **0.98**	0.90	+9%
Steady volume	above **678 cups/day**	556 cups/day	+22%
Forecast band	above **±57%**	±32.5%	+75%
New-hire share	none in 0% new staff to 50% new staff: all-human cheaper throughout		
Machine time	none in 0.70× to 2.00×: all-human cheaper throughout		

Each row moves one input with the rest at our defaults; nearest first. A crossed row is a reason to recheck, not a rule to switch.

How sure are we? A close call. The gap is S$0.02 a cup. Our staffing shortcut misses its Week 5 check (4.7% and 6.8% against ±3%), and with walking modelled people reach only 90% of the assumed capacity (70% at today's counter). Test the ranking with real staffing data before relying on it.

Perform

About 45 seconds; under time pressure, read only the gate card. Speed has a price. The chart is the cost model: labour, machines and lost sales per cup against steady cups per day. At our forecast of about 556 cups a day all-human is cheaper, but only by S$0.02 a cup, so we do not recommend a default. The table says when mixed wins instead, nearest first: machine cost below about S$1,800 a month, all-human staff coordinating a little worse than we assume (below 0.85), or a wage above about S$15.97 an hour; each is only a few per cent away. Volume needs about 678 cups a day. Full automation needs about 1,820. And be clear how sure we are: it is a close call on a simplified staffing model.

Sources: Wage S$ per hour worked, fully loaded: L-mom-pwm, L-mom-ows, L-cpf, L-sdl, L-mom-leave, L-mom-ph. Costs and break-evens: sim/ cost model and breakEven over cups per day (data.json volumeCurve, breakEven), default band, moderate surge. All inputs: assumptions.json (illustrative).

Review: rerun the comparison with observed inputs and compare it with the original plan

GATE 4 · REVIEW

DECISION
After real sales arrive, does the plan still hold?

SUGGESTED ACTION
Create an updated scenario using observed inputs and compare it with the original plan (All-human by S$0.02 a cup).

EVIDENCE
Feeds the model **steady sales · wage · machine quote · milk costs**
Check by hand **service times · lost orders · waste and shortages vs plan**

TRADE-OFF
Change too early and you chase the opening hype; change too late and the store keeps the wrong stock or counter.

LIMITATION
Rerunning the model does not validate it, and a new result does not by itself prove that machines should be added.

TRIGGER	FLIPS WHEN	OURS	GAP
Counter: when mixed beats all-human · Store format / equipment, with franchisees			
Machine cost	below **S$1,800/month**	S$1,909/month	-6%
All-human coordination	below **0.85**	0.90	-6%
Wage per hour	above **S$15.97/h**	S$15.06/h	+6%
Mixed coordination	above **0.98**	0.90	+9%
Steady volume	above **678 cups/day**	556 cups/day	+22%
Forecast band	above **±57%**	±32.5%	+75%
New-hire share, Machine time	no threshold in the tested range: all-human stays cheaper throughout		
Counter: when full automation beats mixed · Store format / equipment, with franchisees			
Steady volume	above **1,820 cups/day** · step	556 cups/day	+228%
Wage per hour, Machine cost, Forecast band, Machine time	no threshold in the tested range: mixed stays cheaper throughout		
Stock: when to order at or below the base forecast · Supply chain / planning			
Milk waste per litre	none in S$1.27 to S$5.09/L: order above base throughout (flips at S$25.47/L = S$3.82 per lost cup ÷ 0.15 L)		

One input at a time, the rest at our defaults; nearest first. Thresholds flag what to recheck, not when to switch (slide 7).

INTERACTIVE EXPLAINER UPDATED SCENARIO VERSUS THE ORIGINAL PLAN

Enter observed inputs. Verdict: the full cost model; under S$0.01 a cup is a tie. Nearest threshold: a live estimate.

Wage per hour		S$15.06
Machine cost, per unit per month		S$1,909
Expected steady cups per day		556
Forecast band (±)		±32.5%
Milk waste cost per litre		S$2.55

Original plan: All-human by S$0.02 a cup. **Updated scenario:** same inputs as the plan.

COUNTER **All-human** cheaper than Mixed by S$0.02 a cup

All-human **S$0.65** Mixed **S$0.66** All-machine **S$0.91**

Nearest threshold (live estimate): Mixed would be cheaper if machine cost, per unit per month falls to about **S$1,783** (7% away).

Stock: order above the base forecast (best order at the 91% point of demand); flips only if waste costs over S$25.47 a litre.

Printed at our default inputs.

Decide

About 50 seconds. This is the call to action. There is no default counter; the table collects every trigger from the simulator, nearest first. For mixed over all-human, three sit within a few per cent of our defaults: machine cost, how well staff coordinate, and the wage. Volume needs about 678 cups a day. Full automation needs about 1,820 cups a day, and nothing else in range gets it there. Stock: order above the base forecast unless milk waste costs over S$25.47 a litre, which is far outside our range. On the right, put in a market: say a higher wage, or a cheaper machine. The panel recomputes the verdict, how much it wins by, and which trigger is closest to flipping it. If the margin is under a cent a cup, call it a tie and decide on other grounds.

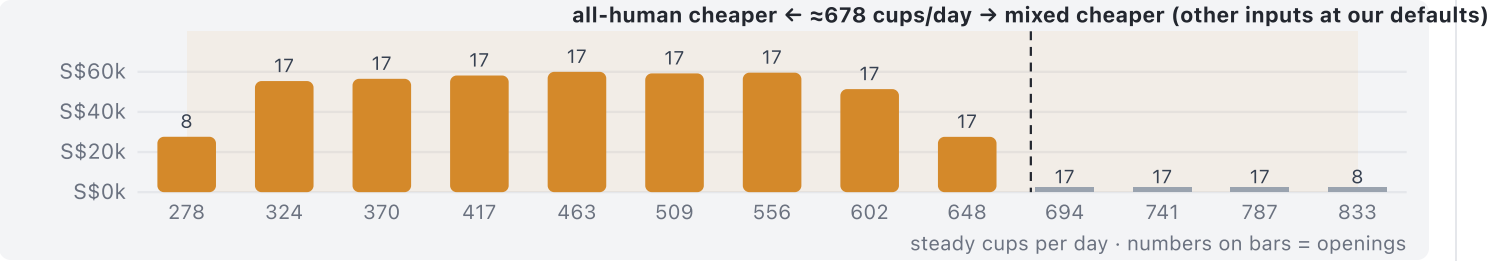
Sources: Wage: L-mom-pwm, L-mom-ows, L-cpf, L-sdl, L-mom-leave, L-mom-ph. Triggers: sim/ breakEven (data.json breakEven, leverNow) and milkTrigger. Your market: sim/ runPoint with simulation.live_draws, nearest flip found by bisection at the user's inputs. All inputs: assumptions.json (illustrative).

Choosing each store's counter for its conditions is worth about S$455k a year in counter costs, plus S$74k once in opening stock (illustrative)

INTERACTIVE EXPLAINER OPENING PIPELINE: WHAT CHOOSING PER STORE IS WORTH

Comparison: a mixed counter at every opening versus choosing a counter for each store's modelled conditions. A modelled avoided cost, not a demonstrated AI saving.

Bar = openings at that volume × the standard's yearly cost above the cheapest counter (our defaults).



Overseas openings a year		200
Expected steady volume: lowest		278 cups/day
Expected steady volume: highest		833 cups/day

RECURRING · COUNTER COSTS
S$455k
a year, for one year's openings

ONE-TIME · OPENING STOCK
S$74k
once, at opening

142 of 200 openings sit where all-human is cheaper → S$455k a year; **S$370** per opening × 200 → S$74k once

Stock term only if the pre-launch signal works as assumed (slide 3). A fully automated standard would cost S$32,724–S$65,190 more per store a year.

Printed at our defaults: openings spread evenly over our volume range.

HOW EACH OPENING IS DECIDED

1. **Before committing the site** · Predict `slide 3`
   Open the waitlist and read the demand band; back-test it on past openings.
   Market-entry leads · Membership / CRM
2. **Before the first order** · Provision `slide 4`
   Order milk at the best order, above the base forecast; that holds unless waste costs over S$25.47 a litre.
   Supply chain / planning
3. **Before fit-out** · Counter `slide 7`
   Compare all three setups with all current inputs together; thresholds only flag what to recheck.
   Store format & franchisee
4. **After opening** · Review `slide 8`
   Build an updated scenario from observed sales, wages and machine quotes, and compare it with the original plan.
   Store operations · Market-entry leads

RISKS AND HOW WE WOULD MANAGE THEM

- **Staff dependence and turnover**
  Keep SOP training and flexible roles.
- **Opening crowds read as steady demand**
  Decide the counter on steady volume, after the hype.
- **Tourist and non-app demand is invisible**
  Widen the band in tourist areas; log walk-ups.
- **Automation versus the premium teahouse**
  Keep people at finishing and hand-over.

FINDING

In the tested model, uncertainty did not determine the counter choice.

Volume and operating costs mattered more; demand uncertainty still matters for inventory. This holds for the scenarios we tested, not as a general rule. Next: replace the assumptions with Chagee data and back-test on recent openings such as Korea (Q2 2026).

Benefit, risks, next step

About 1 minute. This is how an opening is actually decided: four gates. Before committing the site, read the waitlist band. Before the first order, stock milk above the base forecast. Before fit-out, put the store's volume, wage and machine price into the playbook and pick the counter. After opening, measure steady volume and add machine stations at about 678 cups a day. On the left, what that is worth against one standard counter everywhere, Chagee's semi-automated one: move the openings and the expected volume range and the bars follow. Below the trigger the standard costs about S$3,220 a store a year more than the cheapest counter; above it, nothing. Across about 200 openings that is about S$455k a year in counter costs, plus a one-time S$74k of opening stock if the signal works as assumed. A modelled avoided cost, not a demonstrated saving. The risks are on the slide with one mitigation each. Close on the next step: real data, a back-test on recent openings such as Korea, and a paired pilot.

Sources: Overseas openings planned for 2026 and Korea launch: L-jiemian. Semi-automated counter as Chagee's standard: L-unsw, L-cyberrt (our reading). Per-store cost: data.json volumeCurve and the cups-per-day break-even, x open days. Share below the trigger: openings assumed evenly spread over the approved cups range (illustrative). Stock term: provision best-order cost, no signal minus default band. Everything else: assumptions.json and sim/ (illustrative).

Every assumption, with its unit, range and source

ASSUMPTION	VALUE	UNIT	RANGE	SOURCE
Facts				
stores end 2025	7,453	stores	–	L-jiemian
overseas stores	345	stores	–	L-jiemian
overseas markets	7	markets	–	L-jiemian
gmv 2025 rmb bn	31.58	RMB bn	–	L-jiemian
net revenue 2025 rmb bn	12.91	RMB bn	–	L-jiemian
tea latte share china	0.91	share of China sales	–	L-prometheus
kl opening day cups	1,000	cups (reported, '1,000+')	–	L-gabgrowth
Predict				
signal strength default	0.5	share, 0 = no pre-launch signal, 1 = strong signal	0 to 1	assumption
Predict Toggles				
remove hype default	yes	boolean	–	assumption
Store				
open hours per day	12	hours/day	–	assumption
open days per year	360	days/year	–	assumption
cups per store per year	200,000	cups/store/year	100,000 to 300,000	assumption
hourly demand share	0.04, 0.06, 0.1, 0.11, 0.08, 0.08, 0.09, 0.1, 0.1, 0.09, 0.08, 0.07	share of daily cups per open hour (10:00 to 22:00), sums to 1	–	assumption
Demand				
uncertainty definition	half_width_pct	text	–	assumption
band coverage	0.8	probability	–	assumption
demand floor share of base	0	share of base	–	assumption
uncertainty no signal	0.5	half-width, share of base	0.35 to 0.7	assumption
uncertainty strong signal	0.15	half-width, share of base	0.05 to 0.25	assumption
opening hype multiplier	1.5	x steady-state demand	1 to 2.5	assumption
opening hype weeks	2	weeks	1 to 6	assumption
uncertainty sweep	from 0 · to 0.6 · step 0.05	half-width, share of base	–	assumption
Surge				
surge multiplier	none 1 · moderate 1.5 · large 2.5	x forecast arrival rate	–	assumption
surge hours	2	busiest forecast hours that get the surge	–	assumption
surge default	moderate	text	–	assumption

Source "assumption" = our illustrative choice; ledger IDs are listed in A3. Ranges drive the sliders and the sensitivity chart in A2.

Appendix A1 · assumptions

Handout only. Generated from assumptions.json at build time, so it always matches the deck.

Sources: assumptions.json.

Every assumption, with its unit, range and source (continued)

ASSUMPTION	VALUE	UNIT	RANGE	SOURCE
Stations				
list	ticket, cup_ice, tea_milk, topping_finish, shake_seal, handover	text	–	assumption
human seconds per cup	ticket 5 · cup ice 10 · tea milk 15 · topping finish 12 · shake seal 10 · handover 8	seconds of one average worker per cup	0.8 to 1.25	assumption
machine seconds per cup	ticket 2 · cup ice 8 · tea milk 10 · topping finish 15 · shake seal 8 · handover 6	seconds per cup per machine unit	0.7 to 2	assumption
mixed machine stations	ticket, tea_milk, shake_seal	text	–	assumption
machine speed variation cv	0	coefficient of variation	–	assumption
People				
human speed variation cv	0.25	coefficient of variation, minute to minute	0.1 to 0.4	assumption
worker speed spread	0.2	+/- share around average speed	0 to 0.4	assumption
coordination loss human	0.9	multiplier on pooled throughput	0.75 to 1	assumption
coordination loss mixed	0.9	multiplier on human-pool throughput in the mixed setup	0.75 to 1	assumption
new staff share	0.25	share of roster in first months	0 to 0.5	assumption
new staff relative speed	0.7	x average speed	0.5 to 0.9	assumption
min staff on shift	all human 2 · mixed 1 · all machine 0	workers per open hour	–	assumption
all machine attendant per open hour	1	workers per open hour	0.5 to 2	assumption
staffing flexibility	0.7	share of roster gap closed after demand is known	0.3 to 1	assumption
Costs				
labour cost per hour	15.06	SGD per hour worked, fully loaded (statutory costs)	14.09 to 25.45	L-mom-pwm, L-mom-ows, L-cpf, L-sdl, L-mom-leave, L-mom-ph
machine cost per unit per month	1,909	SGD per station machine unit per month	636.2 to 3,818	assumption
lost order cost	3.82	SGD per lost order	1.91 to 7.64	assumption
capacity safety margin	1.2	x forecast peak rate	1 to 1.5	assumption
Service				
target wait minutes	5	minutes	2 to 10	assumption
Provision				
milk litres per cup	0.15	litres per cup	0.1 to 0.25	assumption
milk waste cost per litre	2.55	SGD per litre thrown away	1.27 to 5.09	assumption
milk shelf life days	7	days	3 to 14	assumption
stockout cost per cup	3.82	SGD per cup not served for lack of stock	1.91 to 7.64	assumption
opening stock days	7	days of forecast base demand	1 to 14	assumption
Benefit				
overseas stores per year	200	stores/year	50 to 300	L-jiemian
saving per cup	from_simulator	SGD/cup	–	assumption
waste avoided	from_provision	SGD/store/year	–	assumption

Source "assumption" = our illustrative choice; ledger IDs are listed in A3. Ranges drive the sliders and the sensitivity chart in A2.

Appendix A1 · assumptions (continued)

Handout only. Generated from assumptions.json at build time, so it always matches the deck.

Sources: assumptions.json.

Every assumption, with its unit, range and source (continued)

ASSUMPTION	VALUE	UNIT	RANGE	SOURCE
Simulation				
monte carlo draws	500	demand draws per uncertainty point	–	assumption
base seed	639	integer	–	assumption
minute step	1	minutes	–	assumption
live draws	20	simulated stores per live estimate	–	assumption
Triggers				
volume trigger range cups per day	277.8, 3,000	cups per store per day	–	assumption
Illustration				
worker speeds	1, 1.5, 2	lines per hour	–	assumption
walk back speed	6	lines per hour	–	assumption
base load share	0.6	arrival rate as share of the crew's nominal capacity (sum of speeds)	–	assumption
machine sizing	matched station rates = 3 x worker speeds (each station does a third of the work; same total speed as the cr…	text	–	assumption
mixed sizing	matched machine block rate = the people part's nominal rate (crew speed / hand share), a balanced mixed line…	text	–	assumption
machine buffer capacity	2	orders between consecutive machine stations	–	assumption
run hours	4	simulated hours per run	–	assumption
surge window hours	1, 2	start and end hour of the surge within a run	–	assumption
Validation				
breakeven	tol cost sgd per cup 0.01 · tol x share of range 0.005 · grid points 13	mixed	–	assumption
week5 comparison	tolerance pct 3 · engine Brigade() read from the ENGINE string in week5/build_opim639_deck.py, not modified …	mixed	–	assumption
cost per cup aggregation	ratio_of_sums	text	–	assumption
peak utilisation definition	forecast peak-hour arrival rate x surge multiplier, divided by the setup's nominal capacity in that hour, at…	text	–	assumption
crossover definition	smallest u at which mixed cost/cup <= all-machine cost/cup and stays <= for every larger u in the sweep, lin…	text	–	assumption
crossover stability	seeds 20 · min seeds with crossover 18 · max spread pct points 10	mixed	–	assumption
sensitivity	one-at-a-time: each parameter with a non-null range set to its low then high value, all others at default; r…	text	–	assumption

Source "assumption" = our illustrative choice; ledger IDs are listed in A3. Ranges drive the sliders and the sensitivity chart in A2.

Appendix A1 · assumptions (continued)

Handout only. Generated from assumptions.json at build time, so it always matches the deck.

Sources: assumptions.json.

Every assumption, with its unit, range and source (continued): business case

ASSUMPTION	VALUE	UNIT	RANGE	SOURCE
Business Case				
horizon years	5	years	–	assumption
opening year share	0.5	share of a year a store trades in its opening year	–	assumption
specialist day rate	1,000	SGD per person-day	–	assumption
working days per month	21	days/month	–	assumption
build people	4	people	–	assumption
build months	6	months	–	assumption
integration one time	250,000	SGD, one time	–	assumption
maintenance people	2	full-time people, every year	–	assumption
platform per store month	50	SGD per store in operation per month	–	assumption
review days per opening	1	analyst person-days per new store	–	assumption

Source "assumption" = our illustrative choice; ledger IDs are listed in A3. Ranges drive the sliders and the sensitivity chart in A2.

Appendix A1 · assumptions (continued)

Handout only. Generated from assumptions.json at build time. These cost lines are approved illustrative assumptions and feed only the A4 revenue and cost analysis, never the simulator.

Sources: assumptions.json.

The model is stable across seeds but misses its Week 5 check, and a handful of inputs decide all-human versus mixed

Week 5 check: pooled staffing formula vs the Week 5 engine

WORKER SPEEDS	ENGINE, LINES/H	POOLED	GAP
1, 1.5, 2	4.300	4.500	**4.65%**
1, 2, 3	5.620	6.000	**6.76%**

Fails the pre-registered ±3% tolerance: the pooled formula overstates a real bucket brigade. The coordination factor (0.9) absorbs part of this.

Seed stability

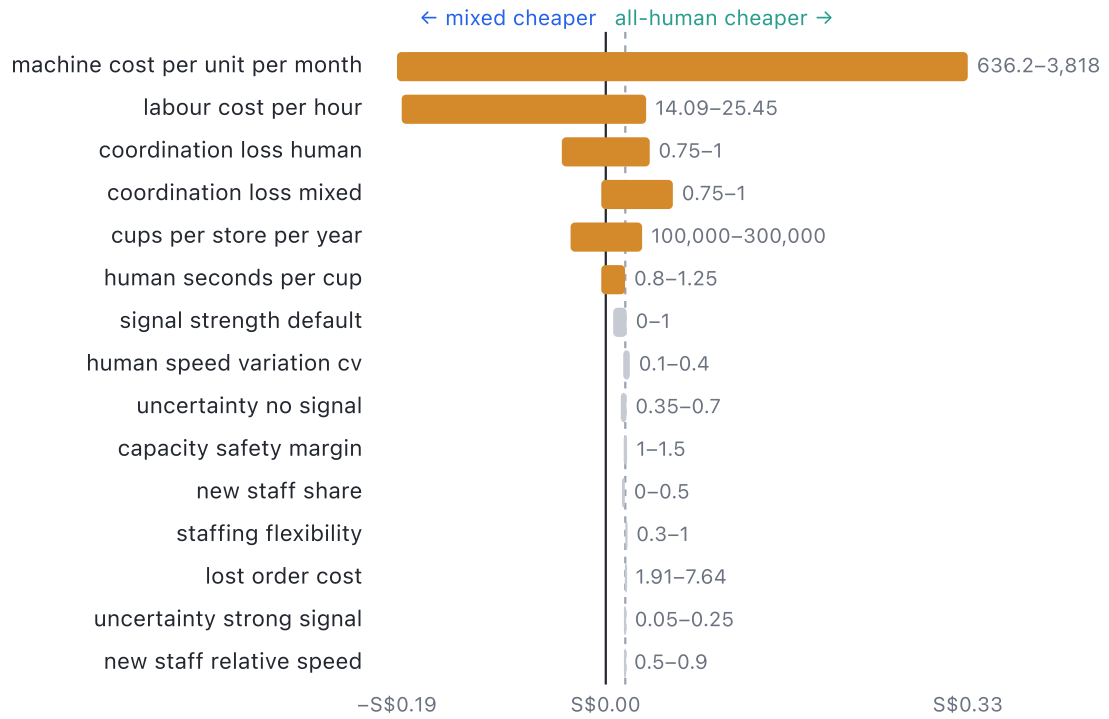
Uncertainty sweep at the moderate surge: 20 of 20 seeds: mixed cheaper than full automation at every band width. The result agrees in every seed. The pre-registered rule asks for a crossover point in most seeds; with none in any seed, the deck shows no crossover point, as the rule requires.

Store replica (walking modelled) vs cost-model capacity, cups per minute

COUNTER	REPLICA	COST MODEL	SHARE
all-human	1.50	1.67	90%
mixed	1.17	1.67	70%
all-machine	3.98	4.00	100%

The replica animates movement only; times and costs on the slides come from the cost model.

Sensitivity: mixed minus all-human cost per cup, one assumption at a time



Dashed: our default gap (S$0.018 a cup, all-human cheaper). Amber bars cross zero inside their approved range, so they can flip the choice; labels show the range tested. No visible effect: target wait minutes, worker speed spread, machine seconds per cup, opening hype multiplier, opening hype weeks, all machine attendant per open hour, milk litres per cup, milk waste cost per litre, milk shelf life days, stockout cost per cup, opening stock days, overseas stores per year.

Appendix A2 · model and validation

Handout only. Week 5 comparison and seed stability from sim/ tests and precompute; the sensitivity chart varies one ranged assumption at a time across its approved range.

Sources: sim/precompute.js: the Week 5 comparison, seed stability, the sensitivity sweep and replica capacity (data.json).

At a new store's volume, a mixed people-and-machine counter beats full automation, because machines sit idle, not because demand is uncertain

We tested the hypothesis that mixed wins only until demand becomes predictable. Our simulator rejected the uncertainty part.

Machines come in whole units, so they sit idle

Even one machine per station serves about 240 cups an hour against a forecast peak of about 61. The fully automated line is busy only 21% to 22% of open hours.

People flex; machines cannot be returned

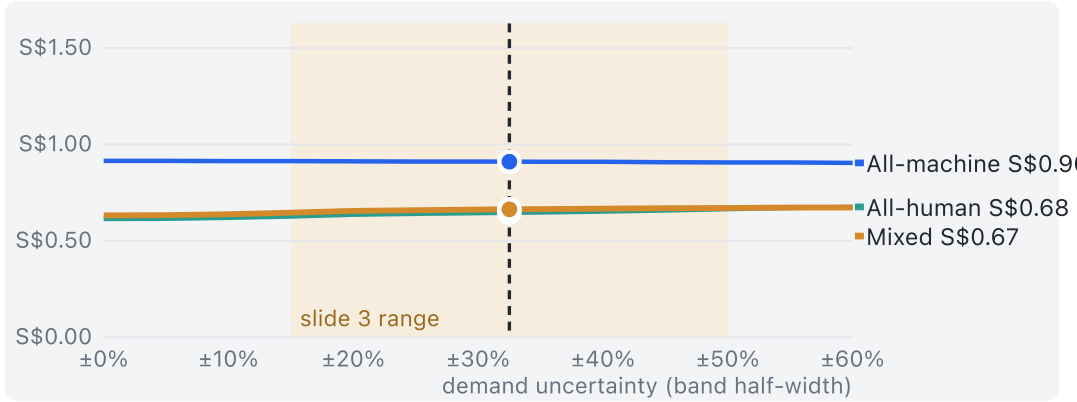
Rosters follow the hour and the real demand, and staff rebalance across stations like a bucket brigade. A mixed counter buys 3 machines instead of 6, with similar staff hours.

Where machines still win

Big surges: at a 2.5× peak, people-based counters lose about 11% of orders and full automation is cheaper. Also steady high volume, consistency and labour scarcity.

INTERACTIVE EXPLAINER COST PER CUP BY COUNTER SETUP

None ×1 surge | **Moderate ×1.5 surge** | Large ×2.5 surge



No crossover: mixed is cheaper than full automation at every uncertainty level shown.

Demand uncertainty ±32.5%

COST PER CUP AT ±32.5%

All-human	S$0.65
Mixed	S$0.66
All-machine	S$0.91
Mixed – machine	−S$0.25

WHAT THE COST IS MADE OF



Automated line busy	21%
Orders lost: mixed	1.3%

What could shift the balance

Machine cost per month	Low S$636	**Default**	High S$3,818	→ mixed cheaper everywhere
Staff turnover (share new)	Low 0%	**Default**	High 50%	→ mixed cheaper everywhere
Forecast narrows (slide 3 band)	No signal ±50%	**Slide 3**	Strong ±15%	

Printed at the default band (±32.5%) and moderate surge.

OUR VIEW

Uncertainty did not decide the counter; volume, wage and machine cost do.

All-human (S$0.65) and mixed (S$0.66 per cup at the default band) are a close call settled by the triggers on slide 7. Machine breakdowns are not modelled, which flatters automation.

Appendix A2 · the uncertainty sweep

Handout only (appendix). Our starting hypothesis was that a mixed counter beats full automation until demand becomes predictable. The simulator only half supports it. Mixed does beat full automation at the default settings, at every level of uncertainty, but the reason is not uncertainty: machines come in whole units, so the automated line sits idle and its fixed cost is spread over too few cups. The uncertainty slider barely moves the lines; a large surge does, and there full automation wins. All-human versus mixed is a close call decided by volume, wage and machine cost (slide 7). Limits: the pooled staffing formula overstates a real bucket brigade (our Week 5 check missed its tolerance, gap 4.7 to 6.8 percent), and breakdowns are not modelled.

Sources: All figures: sim/ with assumptions.json (illustrative). Semi-automation context: L-unsw, L-cyberrt. Not novel - beverage automation: L-gongcha (company-reported), L-slate; store digital twins: L-mosimtec. Labour cost floor: L-mom-pwm, L-cpf.

The source behind every Chagee fact, and the claims we deliberately did not make

ID	SOURCE AND WHAT IT SUPPORTS	STATUS
L-jiemian	Jiemian (en.jiemian.com), earnings-call coverage: 7,453 stores end-2025; about 200 overseas stores planned for 2026; 345 overseas stores; 7 overseas markets. (brie…	PUBLIC
L-gabgrowth	GabGrowth (secondary): first overseas store, Kuala Lumpur, reportedly sold 1,000+ cups on opening day. (brief.md section 9)	SECONDARY
L-unsw	UNSW BusinessThink, Oct 2024: Chagee semi-automation, baristas scan a QR code and the machine dispenses the formulation. (brief.md section 9)	PUBLIC
L-mom-pwm	Ministry of Manpower, Progressive Wage Model, Food Services sector, https://www.mom.gov.sg/employment-practices/progressive-wage-model/food-services-sector (page u…	PUBLIC
L-cpf	CPF Board, 'How much CPF contributions to pay', https://www.cpf.gov.sg/employer/employer-obligations/how-much-cpf-contributions-to-pay (page updated 11 Aug 2026; f…	PUBLIC
L-mom-ows	MOM Occupational Wages 2025, Table 1 (median monthly basic and gross wages of full-time resident employees, June 2025; released 30 Jun 2026), https://stats.mom.gov…	PUBLIC
L-sdl	CPF Board, Skills Development Levy, https://www.cpf.gov.sg/employer/employer-obligations/skills-development-levy (updated 2 Jul 2026): 0.25% of monthly total wages…	PUBLIC
L-mom-leave	MOM, Annual leave eligibility and entitlement, https://www.mom.gov.sg/employment-practices/leave/annual-leave/eligibility-and-entitlement (updated 17 Apr 2023): 7 …	PUBLIC
L-mom-ph	MOM, Public holidays entitlement and pay, https://www.mom.gov.sg/employment-practices/public-holidays-entitlement-and-pay (updated 30 Jul 2025): 11 paid public hol…	PUBLIC
L-prn	PR Newswire, 21 May 2024 (company release): Chagee says it sources all tea and milk directly for domestic operations, with distribution managed through HQ. (brief.…	PUBLIC
L-prometheus	Prometheus Capital Substack (SECONDARY - flag when used): own Yunnan plantations, cold chain to 6,000+ stores; tea lattes about 91% of China sales. (brief.md secti…	SECONDARY
L-chainreact	Chainreact Substack (secondary): founder avoided fruit tea partly for supply chain complexity. (brief.md section 9)	SECONDARY
L-triunity	Studio Triunity, citing the Mixue prospectus: Chagee's pure milk-tea menu seen as simpler to export; Mixue has 7 SEA warehouses and 3 planned local factories. (bri…	PUBLIC
L-cyberrt	Cyber-RT / Citizens Journal, Oct 2022: Chagee Express (Malaysia), semi-automated machine, three drink categories for speed. (brief.md section 9)	PUBLIC
L-gongcha	FastCasual and franchise.org, Jan-Feb 2026 (company-reported): Gong cha Super Wu, up to 65% higher peak-hour productivity, about one minute off prep, stores with a…	PUBLIC
L-starbucks	Constellation Research; Harvard D3: Starbucks Deep Brew - forecasting, scheduling, replenishment, order sequencing; location analytics for new stores. (brief.md se…	PUBLIC
L-mosimtec	MOSIMTEC, Winter Simulation Conference 2025: store digital twins replicate existing layouts. (brief.md section 9)	PUBLIC
L-slate	Slate, Aug 2021: bubble tea shaker robot needed about one worker vs four to six; two earlier all-robot shops closed within about a year. (brief.md section 9)	PUBLIC
I-leeyen	INFORMAL - call with Lee Yen. Do not cite without checking with her: Yunnan standard sourcing, GM-led new-market forecasting, app-only ordering with tourist except…	INFORMAL
L-xrates	X-Rates monthly averages, https://x-rates.com/average/?amount=1&from=USD&to=SGD&year=2026 (fetched 4 Oct 2026). September 2026 USD to SGD average 1.272467 (30 days…	SECONDARY

Claims we did not make

- That Chagee is first at anything, or that anything is not available in the market.
- Any revenue-mix share for franchise supply: sources conflict.
- Overseas milk sourcing details: unknown, so milk costs are generic.
- A Supply Chain Security Act local-sourcing rule: low-quality source.
- Any Chagee internal cost, forecast method or automation plan not in the ledger.
- Measured serve times: every time on the store-replica slide is modelled; the observed row is empty.
- That member data narrows the forecast band: the narrowing is our assumption, tested as a sensitivity.
- That all-human or mixed is proven cheaper: the gap is small and rests on a simplified staffing model.
- A saving against today's practice: the benefit is a modelled avoided cost against a standard counter.
- Anything about menu prices: that would need a demand-elasticity assumption we do not have.
- The informal sub-one-minute fulfilment target from the industry call: not used.
- That the prototype contains a trained AI model: it calculates from synthetic, assumed inputs.
- That rerunning the model with observed inputs validates it, or proves machines should be added.
- That uncertainty never matters for the counter: the finding holds for the scenarios we tested.
- Sales revenue or a payback for Chagee: A4 sets modelled avoided cost against approved but illustrative costs.

Appendix A3 · sources and claims not made

Handout only. Ledger from assumptions.json _sources; the list of claims not made follows brief.md section 9 plus the limits found while building the deck.

Sources: assumptions.json _sources; brief.md section 9.

Over 5 years the modelled savings cover the cost of building and running the gates in year 5, leaving about S$289k ahead (illustrative)

ITEM	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	ASSUMPTION / HOW CALCULATED
Benefit: modelled avoided cost (no sales revenue)						
New stores opened	200	200	200	200	200	About 200 overseas openings planned for 2026 (L-jiemian); held flat after that (no later figure in our sources)
Store-years using the gates	100	300	500	700	900	Earlier stores trade all year; new stores 50% of a year (mid-year openings)
Counter saving (recurring)	S$228k	S$683k	S$1,138k	S$1,594k	S$2,049k	Store-years × S$2,277: the mixed counter's yearly cost above the cheapest counter, openings spread over our volume range (slide 9)
Opening-stock saving (once per opening)	S$74k	S$74k	S$74k	S$74k	S$74k	Openings × S$370: best milk order with no signal minus with the member signal; holds only if the signal narrows the range as assumed
Total benefit	**S$302k**	**S$757k**	**S$1,213k**	**S$1,668k**	**S$2,123k**	
Cost of the initiative: one-time						
Build the four gates	S$504k	–	–	–	–	4 people × 6 months × 21 days × S$1,000 a day: forecast, ordering, counter comparison, review
Data integration and set-up	S$250k	–	–	–	–	POS sales, app sign-ups and supplier order feeds; lump sum
Total one-time cost	**S$754k**	**–**	**–**	**–**	**–**	
Cost of the initiative: recurring						
Maintenance team	S$504k	S$504k	S$504k	S$504k	S$504k	2 people × 12 months × 21 days × S$1,000; counted from year 1 (conservative)
Platform and licences	S$60k	S$180k	S$300k	S$420k	S$540k	Store-years × S$50 a month × 12
Gate 4 reviews	S$200k	S$200k	S$200k	S$200k	S$200k	Openings × 1 analyst day × S$1,000: updated scenario against the original plan
Total recurring cost	**S$764k**	**S$884k**	**S$1,004k**	**S$1,124k**	**S$1,244k**	
Total cost	**S$1,518k**	**S$884k**	**S$1,004k**	**S$1,124k**	**S$1,244k**	
Net benefit	**(S$1,216k)**	**(S$127k)**	**S$209k**	**S$544k**	**S$879k**	**Total benefit minus total cost**
Cumulative net	**(S$1,216k)**	**(S$1,343k)**	**(S$1,135k)**	**(S$591k)**	**S$289k**	**From year 1**

Sensitivity, cumulative net after 5 years. one-time costs double: S$465k short, no payback; counter saving halves: S$2,558k short, no payback; openings fall to 100 a year: S$1,493k short, no payback.

Read with care. No sales revenue: we have no price or demand-elasticity assumption, so the benefit is the modelled avoided cost from slide 9, against a mixed counter at every opening; it is not a demonstrated AI saving. The cost lines are our approved illustrative assumptions (A1, business case), not Chagee or vendor figures. Values in S$ thousands; brackets are negative.

Appendix A4 · revenue and cost analysis

Handout only. Benefit lines come from the benefit slide's calculation (sim/ volumeCurve and provision); cost lines are the approved business-case assumptions (illustrative). No sales revenue is modelled.

Sources: Openings: L-jiemian (2026 only; held flat after). Benefit: benefit_values in deck/build_deck.py from data.json. Costs: assumptions.json business_case (approved, illustrative).